

1. MC.05.01.ALGO (Algorithmic)

A batch-level driver is consumed by a product each and every time that

- ✓

☒

a. a batch of products is produced.
- ☐

b. a finished unit is sold.
- ☐

c. a customer complains.
- ☐

d. a unit is produced.
- ☐

e. None of these.

Feedback

Check My Work

Activities are often classified as unit-level, batch-level, product-level, and facility-level.

Solution
a

2. MC.05.02.ALGO (Algorithmic)

Which of the following is a nonunit-level driver?

- ✓

☒

a. Setup hours
- ☐

b. Indirect materials
- ☐

c. Assembly hours
- ☐

d. Direct materials
- ☐

e. Direct labor hours

Feedback

Check My Work

Nonunit activities are not performed each time a unit is produced.

Solution
a

3. MC.05.03

Consider the information given on two products and their activity usage:

	Laser Printer	Inkjet Printer
Units produced	1,000	4,000
Setup hours	800	400
Inspection hours	500	500
Machine hours	200	1,000

The consumption ratios for the inspection activity for each product are:

- ✓

☒

c. 0.500; 0.500.
- ☐

a. 0.167; 0.833.
- ☐

b. 0.333; 0.667.
- ☐

d. 0.667; 0.333.
- ☐

e. None of these.

Feedback

Check My Work

Identify the activity driver for each activity.

Divide the amount of driver used for each product by the total driver quantity.

Post-Submission

Amount of activity driver per product / Total driver quantity

$$500/1,000 = 0.500; 500/1,000 = 0.500$$

Solution

c

4. MC.05.04

Consider the information given on two products and their activity usage:

	Laser Printer	Inkjet Printer
Units produced	1,000	4,000
Setup hours	800	400
Inspection hours	500	500
Machine hours	200	1,000

Suppose that machine hours are used to assign all overhead costs to the two products. Select the best answer from the following:

- 
- ☐ a. Laser printers are overcosted, and inkjet printers are undercosted.
 - ☐ b. Laser printers and inkjet printers are accurately costed.
 - ☒ c. Laser printers are undercosted, and inkjet printers are overcosted.
 - ☐ d. Using inspection hours to assign overhead costs is the most accurate approach.
 - ☐ e. None of these.

Feedback

Check My Work

Look at what drives overhead costs. Is it only one driver? Or are more drivers needed to assign costs accurately?

Post-Submission

Setup hours are twice as much for Laser, consuming more overhead.

Solution

c

5. MC.05.05.ALGO (Algorithmic)

The first stage of ABC entails the assignment of

- ☐ a. resource costs to distribution channels.
- ☐ b. activity costs to product sales units.

- ☐ c. resource costs to departments.
- ☐ d. resource costs to a plantwide pool.
- ✓ ☒ e. resource costs to individual activities.

Feedback

Check My Work

The first stage of overhead costing traces costs to an intermediate object. The second stage traces the costs collected in the first stage to the final cost object.

Solution

e

6. MC.05.06

The second stage of ABC entails the assignment of

- ✓ ☒ a. activity costs to products or customers.
- ☐ b. resource costs to departments.
- ☐ c. resource costs to a plantwide pool.
- ☐ d. resource costs to individual activities.
- ☐ e. resource costs to distribution channels.

Feedback

Check My Work

The first stage of overhead costing traces costs to an intermediate object. The second stage traces the costs collected in the first stage to the final cost object.

Solution

a

7. MC.05.07

Interview questions are asked to determine

- ☐ a. what activities are being performed.
- ☐ b. who performs the activities.
- ☐ c. the relative amount of time spent on each activity by individual workers.
- ☐ d. possible activity drivers for assigning costs to products.
- ✓ ☒ e. All of these.

Feedback

Check My Work

Interviews are needed to collect information needed for implementing an activity-based accounting system.

Solution

e

8. MC.05.08.ALGO (Algorithmic)

The receiving department employs one worker who spends 75% of his time on the receiving activity and 25% of his time on inspecting products. His salary is \$40,000. The amount of cost assigned to the receiving activity is

- ✓ ☒ a. \$30,000.
- ☐ b. \$10,000.

- ☐ c. \$40,000.
- ☐ d. \$34,000.
- ☐ e. None of these.

Feedback

Check My Work

Resource drivers or their consumption ratios are used to assign the costs of resources to activities.

Post-Submission

$$\$40,000 \times 0.75 = \$30,000$$

Solution

a

9. MC.05.09

Assume that the moving activity has an expected cost of \$80,000. Expected direct labor hours are 20,000, and expected number of moves is 40,000. The best activity rate for moving is

- ☐ a. \$4 per move.
- ☐ b. \$1.33 per hour-move.
- ☐ c. \$4 per hour.
- ☒ d. \$2 per move.
- ☐ e. None of these.



Feedback

Check My Work

The driver that has the best cause-and-effect relationship should be used.

Post-Submission

$$\$80,000/40,000 \text{ moves} = \$2 \text{ per move}$$

Number of moves is the best driver.

Solution

d

10. MC.05.10

Which type of greenhouse gas emission is defined as arising from a company's direct operations, such as an on-site power plant production of electricity?

- ☒ a. Scope 1
- ☐ b. Scope 2
- ☐ c. Scope 3
- ☐ d. All of the options
- ☐ e. None of these options



Feedback

Check My Work

Correct

Solution

a

11. MC.05.11.ALGO (Algorithmic)

Which of the following is a true statement about activity-based customer costing?

- 
- ☒ a. Customer diversity requires multiple drivers to trace costs accurately to customers.
- ☐ b. It is considered to be part of financial accounting activities.
- ☐ c. It seldom produces changes in the company's customer mix.
- ☐ d. It guarantees that the firm will earn a profit from serving each of its customers.
- ☐ e. None of these.

Feedback

Check My Work

ABC is used to more accurately determine upstream costs of suppliers and downstream costs of customers. Knowing the costs of customers can be vital information for improving a company's profitability.

Solution
a

12. MC.05.12

Which of the following is a true statement about activity-based supplier costing?

- 
- ☐ a. The cost of a supplier is the purchase price of the components or materials acquired.
- ☐ b. It encourages managers to increase the number of suppliers.
- ☐ c. It encourages managers to evaluate suppliers based on purchase cost.
- ☒ d. Suppliers can affect many internal activities of a firm and significantly increase the cost of purchasing.
- ☐ e. All of these.

Feedback

Check My Work

ABC is used to more accurately determine upstream costs of suppliers and downstream costs of customers. Knowing the costs of suppliers can be vital information for improving a company's profitability.

Solution
d

13. MC.05.13

This year, Lambert Company will ship 1,500,000 pounds of goods to customers at the cost of \$1,200,000. If a customer orders 10,000 pounds and produces \$200,000 of revenue (total revenue is \$20 million), the amount of shipping cost assigned to the customer by using ABC would be

- 
- ☐ a. unable to be determined.
- ☒ b. \$8,000 (\$0.80 per pound shipped).
- ☐ c. \$24,000 (2% of the shipping cost).
- ☐ d. \$12,000 (1% of the shipping cost).
- ☐ e. None of these.

Feedback

Check My Work

You need to select the best driver to assign the shipping costs—the one that has the stronger causal relationship.

Post-Submission

$\$1,200,000/1,500,000 \text{ pounds} = \$0.80 \text{ per pound shipped}$

Order of 10,000 pounds $\times 0.80 = \$8,000$ for shipping

Solution
b

14. MC.05.14

Lambert Company has two suppliers: Deming and Leming. The cost of warranty work due to defective components is \$2,000,000. The total units repaired under warranty average 100,000, of which 90,000 have components from Deming and 10,000 have components from Leming. Select the items below that represent true statements.

- 
- ☐ a. Components purchased from Leming cost \$200,000 more than their purchase price.
 - ☐ b. Components purchased from Deming cost \$1,800,000 more than their purchase price.
 - ☐ c. Components from Leming appear to be of higher quality.
 - ☒ d. All of these.
 - ☐ e. None of these.

Feedback

Check My Work

Determine cost per repaired unit. Multiply that by repaired units for both companies. Compare the warranty cost for both companies' products.

Post-Submission

$\$2,000,000/100,000 \text{ units repaired} = \20 per repair

Leming repairs $10,000 \times \$20 = \$200,000$

Deming repairs $90,000 \times \$20 = \$1,800,000$

Components from Leming appear to be of higher quality because there are fewer repairs for units with Leming components.

Solution
d

15. MC.05.15

A forklift and its driver used for moving materials are examples of

- 
- ☒ a. activity inputs.
 - ☐ b. activity output measures.
 - ☐ c. resource drivers.
 - ☐ d. activity outputs.
 - ☐ e. root causes.

Feedback

Check My Work

Activity inputs are resources consumed by the activity. Activity output is the result or product of an activity. Activity output measures the number of times the activity is performed. Driver analysis is done to determine root causes of activities. Resource drivers measure the quantity of resources consumed by an activity.

Solution
a

16. MC.05.16

Which of the following are nonvalue-added activities?

- ☐ a. Moving goods
- ☐ b. Storing goods
- ☐ c. Inspecting finished goods
- ☐ d. Reworking a defective product
- ☒ e. All of these.



Feedback

Check My Work

Activities necessary to remain in business are necessary activities and are value-added. Three conditions must be met to qualify: the activity produces a change of state, the change of state was not achievable by previous activities, and the activity enables other activities to be performed. Any activity not meeting all three criteria is classified as nonvalue-added.

Solution
e

17. MC.05.17

Suppose that a company is spending \$60,000 per year for inspecting, \$30,000 for purchasing, and \$40,000 for reworking products. A good estimate of nonvalue-added costs would be

- ☐ a. \$70,000.
- ☐ b. \$130,000.
- ☐ c. \$40,000.
- ☐ d. \$90,000.
- ☒ e. \$100,000.



Feedback

Check My Work

Activities necessary to remain in business are necessary and are value-added. Three conditions must be met to qualify: the activity produces a change of state, the change of state was not achievable by previous activities, and the activity enables other activities to be performed. Any activity not meeting all three criteria is classified as nonvalue-added.

Post-Submission

Inspecting and reworking are non-value added activities.

$$\$60,000 + \$40,000 = \$100,000$$

Purchasing is a value-added activity.

Solution
e

18. MC.05.18

The cost of inspecting incoming parts is most likely to be reduced by

- ☐ a. activity sharing.
- ☒ b. activity elimination.
- ☐ c. activity reduction.
- ☐ d. activity selection.
- ☐ e. None of these.



Check My Work

Activity management handles cost reduction with activity elimination, reduction, sharing, or selection.

Post-Submission

Inspecting incoming parts is necessary only because of the poor quality performance of the supplying firms. Selecting suppliers who are able to supply high-quality parts or who are willing to improve their quality of performance to achieve this objective will eventually allow the elimination of incoming inspection.

Solution

b

19. MC.05.19

Thom Company produces 60 units in 10 hours. The cycle time for Thom

- ✓
- ☐ a. is 6 units per hour.
 - ☐ b. is 10 hours per unit.
 - ☒ c. is 10 minutes per unit.
 - ☐ d. is 6 minutes per unit.
 - ☐ e. cannot be calculated.

Feedback

Check My Work

Cycle time measures how long it takes to produce an output from start to finish.

Post-Submission

$(10 \text{ hours}/60 \text{ units}) \times 60 \text{ minutes in } 1 \text{ hours} = 10 \text{ minutes}$

Solution

c

20. MC.05.20

Thom Company produces 60 units in 10 hours. The velocity for Thom

- ✓
- ☒ a. is 6 units per hour.
 - ☐ b. is 10 hours per unit.
 - ☐ c. is 10 minutes per unit.
 - ☐ d. is 6 minutes per unit.
 - ☐ e. cannot be calculated.

Feedback

Check My Work

Velocity is the number units of output produced in a given time period.

Post-Submission

$60 \text{ units}/10 \text{ hours} = 6 \text{ units per hour}$

Solution

a

21. MC.05.21

Striving to produce the same activity output with lower costs for the input used is concerned with which of the following dimensions of activity performance?

- 
- ☐

a. Quality
- ☐

b. Time
- ☐

c. Activity sharing
- ☐

d. Effectiveness
- ☒

e. Efficiency

Feedback

Check My Work

The three dimensions of measures for activity performance are efficiency, time, and quality. Consider how each is usually measured.

Post-Submission

Efficiency focuses on the relationship of activity inputs to activity outputs.

Solution
e

22. MC.05.22

The total activity-based cost per unit of a product is \$15.00. After an analysis of all activities, it was discovered that of the \$15 per unit, \$10.00 is value-added and \$5.00 is nonvalue-added. The data analytic models that produced the \$15.00, \$10.00, and \$5.00 are using which data analytic types?

- 
- ☐

a. Predictive, descriptive, and diagnostic
- ☐

b. Descriptive and prescriptive
- ☒

c. Prescriptive, descriptive, and diagnostic
- ☐

d. Descriptive only

Feedback

Check My Work
Correct

Post-Submission

The \$15 is certainly *descriptive*; the analysis that classifies activities and allows the calculation of the value-added and nonvalue-added unit costs is *diagnostic*; finally, the \$5.00 and \$10.00 are costs that should happen and are, therefore, *prescriptive*.

Solution
c

23. MC.05.23

Lean manufacturing seeks to achieve all of the following except for

- 
- ☐

a. zero defects.
- ☐

b. maximizing customer value.
- ☐

c. zero inventories.
- ☒

d. waiting.

Feedback

Check My Work
Correct

Solution
d

24. MC.05.24

When materials costs differ significantly among products in a value stream, the value-stream product cost is calculated by which of the following?



- ☐ a. Unit Materials Cost + Value-Stream Costs/Units Shipped
- ☒ b. Unit Materials Cost + Value-Stream Conversion Costs/Units Shipped
- ☐ c. Value-Stream Costs/Units Produced
- ☐ d. Value-Stream Costs/Units Shipped

Feedback

Check My Work
Correct
Solution
b

25. MC.05.25

A manufacturing cell within a value stream has three processes and the following associated processing times:

Drilling: 12 minutes
Inserting: 6 minutes
Finishing: 2 minutes

How many units can the cell produce per hour (on a continuous running basis)?



- ☐ a. 10 units per hour
- ☐ b. 3 units per hour
- ☒ c. 5 units per hour
- ☐ d. 30 units per hour

Feedback

Check My Work
Correct
Post-Submission
(60 minutes per hour/12 minutes for drilling)
Solution
c

26. PR.05.174.ALGO (Algorithmic)

Tires Unlimited Corp. produces two types of tires: regular and snow tires. The company uses activity-based costing and has identified two activities and their budgeted costs:

Molding \$72,500
Assembly 284,000

Molding is based on mixing hours, and assembly is based on machine hours. Information regarding each product is as follows:

	Regular Tire	Snow Tire
Units produced	5,300	2,200
Mixing hours	595	245

Machine hours 1,040 410

Required:

If required, round answers to two decimal places or the nearest cent. When required, use rounded amounts in subsequent computations.

A. Calculate the activity rates for molding and assembly.

Molding \$ 

86.31

Assembly \$ 

195.86

B. Calculate the total overhead assigned to each type of tire.

Regular \$ 

255048.85

Snow \$ 

101448.55

C. Calculate the unit overhead assigned to each type of tire.

Regular unit cost \$ 

48.12

Snow unit cost \$ 

46.11

D. Assume the company does not use activity-based costing. Calculate the plantwide overhead rate, based on machine hours.

\$ 

245.86

E. Calculate the total overhead assigned to each type of tire using the plantwide overhead rate.

Regular \$ 

255694.40

Snow \$ 

100802.60

Feedback

Check My Work

Correct

Post-Submission

A. Molding $\$72,500 \div 840 = \86.31

Assembly $\$284,000 \div 1,450 = \195.86

B. Regular

$\$86.31 \times 595 =$ $\$51,354.45$

$\$195.86 \times 1,040 =$ $203,694.40$

\$255,048.85

Snow

$$\$86.31 \times 245 = \$21,145.95$$

$$\$195.86 \times 410 = 80,302.60$$

$$\underline{\underline{\$101,448.55}}$$

C. Unit cost**Regular**

$$255,048.85 \div 5,300 = \$48.12$$

Snow

$$\$101,448.55 \div 2,200 = \$46.11$$

D. Plantwide overhead rate:

$$\$356,500 \div 1,450 = \$245.86$$

E.**Regular:**

$$\$245.86 \times 1,040 = \$255,694.40$$

Snow:

$$\$245.86 \times 410 = \$100,802.60$$

Solution

Tires Unlimited Corp. produces two types of tires: regular and snow tires. The company uses activity-based costing and has identified two activities and their budgeted costs:

Molding \$72,500

Assembly 284,000

Molding is based on mixing hours, and assembly is based on machine hours. Information regarding each product is as follows:

	Regular Tire	Snow Tire
Units produced	5,300	2,200
Mixing hours	595	245
Machine hours	1,040	410

Required:

If required, round answers to two decimal places or the nearest cent. When required, use rounded amounts in subsequent computations.

A. Calculate the activity rates for molding and assembly.

Molding \$ Assembly \$

B. Calculate the total overhead assigned to each type of tire.

Regular \$

Snow \$ 101448.55

C. Calculate the unit overhead assigned to each type of tire.

Regular unit cost \$ 48.12

Snow unit cost \$ 46.11

D. Assume the company does not use activity-based costing. Calculate the plantwide overhead rate, based on machine hours.

\$ 245.86

E. Calculate the total overhead assigned to each type of tire using the plantwide overhead rate.

Regular \$ 255694.40

Snow \$ 100802.60

27. PR.05.175.ALGO (Algorithmic)

Randy Company produces candy sticks (hooked and straight). Both products pass through two producing departments. The hooked sticks' production is much more labor intensive than the straight sticks. The straight stick is most popular. The following data have been gathered for the two products:

Product Data Product Data

Hooked

Straight

Units produced per year	242,000	2,150,000
Machine hours	20,000	41,500
Packing orders	22,500	27,000

Randy has decided to use activity-based costing and has identified two types of costs, machine-related (based on machine hours) and other overhead (based on packing orders). Machine-related costs equal \$146,000. Other overhead costs equal \$274,000.

Required:

If required, round your answers to three decimal places. When required, use rounded amounts in subsequent computations.

A. Calculate the activity rate for machine-related costs based on machine hours.

\$ ✓

2.374

per machine hour

B. Calculate the activity rate for other overhead costs based on packing orders.

\$ ✓

5.535

per packing order

C. Using the two activity rates, calculate the overhead cost per unit for hooked sticks.

\$ ✓

0.711

D. Using the two activity rates, calculate the overhead cost per unit for straight sticks.

\$ ✓

0.115

A. Machine-Related Rate = $\$146,000 \div (20,000 + 41,500) = \2.374 per machine hour

B. Other Overhead Rate = $\$274,000 \div (22,500 + 27,000) = \5.535 per packing order

C. Overhead Cost per Unit for Hooked Sticks = $[(\$2.374 \times 20,000) + (\$5.535 \times 22,500)] \div 242,000 = \0.711

D. Overhead Cost per Unit for Straight Sticks = $[(\$2.374 \times 41,500) + (\$5.535 \times 27,000)] \div 2,150,000 = \0.115

Solution

Randy Company produces candy sticks (hooked and straight). Both products pass through two producing departments. The hooked sticks' production is much more labor intensive than the straight sticks. The straight stick is most popular. The following data have been gathered for the two products:

	Product Data Hooked	Product Data Straight
Units produced per year	242,000	2,150,000
Machine hours	20,000	41,500
Packing orders	22,500	27,000

Randy has decided to use activity-based costing and has identified two types of costs, machine-related (based on machine hours) and other overhead (based on packing orders). Machine-related costs equal \$146,000. Other overhead costs equal \$274,000.

Required:

If required, round your answers to three decimal places. When required, use rounded amounts in subsequent computations.

A. Calculate the activity rate for machine-related costs based on machine hours.

\$ per machine hour

B. Calculate the activity rate for other overhead costs based on packing orders.

\$ per packing order

C. Using the two activity rates, calculate the overhead cost per unit for hooked sticks.

\$

D. Using the two activity rates, calculate the overhead cost per unit for straight sticks.

\$

28. PR.05.176.ALGO (Algorithmic)

Thompson Company uses activity-based costing to determine product cost. Three activities and their rates have been calculated as follows:

Setting up equipment	\$520 per setup
Moving goods	\$30 per move
Machining	\$4 per machine hour

Thompson provided the following data from the job-order cost sheet for Job #345:

Direct materials	\$2,800
Direct labor	2,100
Setups	1
Moves	20
Machine hours	1,000

Required:

A. Calculate the overhead applied to Job #345.

\$ 

5120

B. Calculate the total cost of Job #345.

\$ 

10020

C. If price is calculated by applying a 30% markup on cost, what is the price for Job #345?

\$ 

13026

D. Assume Job #345 required two setups, 12 moves, and 600 machine hours. Calculate the overhead applied to Job #345.

\$ 

3800

E. Assuming a 50% markup on cost, what is the price of Job 345 using the overhead rate calculated in part (D)?

\$ 

13050

Feedback

Check My Work
Correct
Post-Submission

A. Overhead Applied = $(\$520 \times 1) + (\$30 \times 20) + (\$4 \times 1,000) = \$5,120$

B.

Direct materials	\$2,800
Direct labor	2,100
Applied overhead	5,120
Total cost	<u><u>\$10,020</u></u>

C. Price = $\$10,020 \times 1.30 = \$13,026$

D. Overhead Applied = $(\$520 \times 2) + (\$30 \times 12) + (\$4 \times 600) = \$3,800$

E. Price = $(\$2,800 + \$2,100 + \$3,800) \times 1.50 = \$13,050$

Solution

Thompson Company uses activity-based costing to determine product cost. Three activities and their rates have been calculated as follows:

Setting up equipment	\$520 per setup
Moving goods	\$30 per move
Machining	\$4 per machine hour

Thompson provided the following data from the job-order cost sheet for Job #345:

Direct materials	\$2,800
Direct labor	2,100
Setups	1
Moves	20
Machine hours	1,000

Required:

A. Calculate the overhead applied to Job #345.

\$ 5120

B. Calculate the total cost of Job #345.

\$ 10020

C. If price is calculated by applying a 30% markup on cost, what is the price for Job #345?

\$ 13026

D. Assume Job #345 required two setups, 12 moves, and 600 machine hours. Calculate the overhead applied to Job #345.

\$ 3800

E. Assuming a 50% markup on cost, what is the price of Job 345 using the overhead rate calculated in part (D)?

\$ 13050

29. PR.05.177.ALGO (Algorithmic)

Assume that the accounts payable department of a company has five clerks; each one is paid \$21,500 per year (\$156,500 total clerical salaries). On average, the clerks spend the following:

- 20% of their time on processing payments
- 30% of their time matching invoices, receiving documents, and billing statements
- 50% of their time correcting errors in the various documents

In addition, long-distance telephone costs of \$1,000 are directly traced to the activity "correcting errors."

Required:

A. Assign the clerical salary cost to each of the three activities.

Processing Payments	\$ 31300	✓
Matching Documents	\$ 46950	✓
Correcting Errors	\$ 78250	✓

B. What is the total cost of the "correcting errors" activity?

\$ 

79250

Feedback

Check My Work
Correct
Post-Submission

A. Processing Payments = $0.20 \times \$156,500 = \$31,300$
Matching Documents = $0.30 \times \$156,500 = \$46,950$
Correcting Errors = $0.50 \times \$156,500 = \$78,250$

B. Correcting Errors = $\$78,250 + \$1,000 = \$79,250$

Solution

Assume that the accounts payable department of a company has five clerks; each one is paid \$21,500 per year (\$156,500 total clerical salaries). On average, the clerks spend the following:

- 20% of their time on processing payments
- 30% of their time matching invoices, receiving documents, and billing statements
- 50% of their time correcting errors in the various documents

In addition, long-distance telephone costs of \$1,000 are directly traced to the activity "correcting errors."

Required:

A. Assign the clerical salary cost to each of the three activities.

Processing Payments \$	31300
Matching Documents \$	46950
Correcting Errors \$	78250

B. What is the total cost of the "correcting errors" activity?

\$

30. PR.05.178.ALGO (Algorithmic)

Assume that the accounts payable department of a company has 10 clerks; each one is paid \$25,000 per year (\$105,500 total clerical salaries). Six of the clerks spend the following percentages of time on the three activities:

- 20% of their time on processing payments
- 30% of their time matching invoices, receiving documents, and billing statements
- 50% of their time correcting errors in the various documents

One clerk, the head clerk, spends 50% of their time on administrative duties and the remainder on error correction. The remaining three clerks spend 30% of their time processing payments and the other 70% of their time on matching documents.

Long-distance telephone costs of \$1,100 are directly traced to the activity "correcting errors."

Required:

A. Assign the clerical salary cost to each of the three activities and to administration.

	Processing Payments	Matching Documents	Correcting Errors	Administration
Clerks 1–6	\$ ✓ 30000	\$ ✓ 45000	\$ ✓ 75000	
Clerks 7–9	✓ 22500	✓ 52500		
Head clerk			✓ 12500	\$ ✓ 12500
Total cost	\$ ✓ 52500	\$ ✓ 97500	\$ ✓ 87500	\$ ✓ 12500

B. What is the total cost of the "correcting errors" activity?

\$ ✓

88600

Feedback

Check My Work

Correct

Post-Submission

B. Total Cost of Correcting Errors = \$87,500 + \$1,100 = \$88,600

Solution

Assume that the accounts payable department of a company has 10 clerks; each one is paid \$25,000 per year (\$105,500 total clerical salaries). Six of the clerks spend the following percentages of time on the three activities:

- 20% of their time on processing payments
- 30% of their time matching invoices, receiving documents, and billing statements
- 50% of their time correcting errors in the various documents

One clerk, the head clerk, spends 50% of their time on administrative duties and the remainder on error correction. The remaining three clerks spend 30% of their time processing payments and the other 70% of their time on matching documents.

Long-distance telephone costs of \$1,100 are directly traced to the activity "correcting errors."

Required:

A. Assign the clerical salary cost to each of the three activities and to administration.

	Processing Payments	Matching Documents	Correcting Errors	Administration
Clerks 1–6	\$ 30000	\$ 45000	\$ 75000	
Clerks 7–9	22500	52500		
Head clerk			12500	\$ 12500
Total cost	\$ 52500	\$ 97500	\$ 87500	\$ 12500

B. What is the total cost of the "correcting errors" activity?

\$ 88600

Armstrong Company produces a variety of bicycles. One of its plants produces two bicycles: a mountain model and a racing model. At the beginning of the year, the following data were prepared for this plant:

	Mountain	Racing
Quantity	281,500	125,500
Selling price per unit	\$1,400	\$1,100
Unit Prime Cost	\$450	\$600

In addition, the following information was provided so that overhead costs could be assigned to each product:

Activity Name	Activity Driver	Activity Cost	Mountain	Racing
Machining	Machine hours	\$22,660,000	262,500	277,000
Engineering	Engineering hours	8,652,000	79,000	57,000
Packing	Packing orders	3,304,000	57,500	124,500

Armstrong Company uses activity-based costing to calculate product costs.

Required:

If required, round each activity driver to the nearest cent. Use rounded amounts in subsequent computations, if applicable.

A. Calculate the per-unit product cost for a mountain bike.

\$

510.73

B. Calculate the per-unit product cost for a racing bike.

\$

739.61

C. Assuming Armstrong Company adds 40% to the cost of a mountain bike and 35% to the cost of a racing bike, calculate the selling prices for a mountain bike and a racing bike. Round "Unit product costs" calculations to 3 decimal places and round your final answers to the nearest whole dollar.

Mountain Bike \$

715

Racing Bike \$

998

Feedback

Check My Work
Correct
Post-Submission

A. Mountain Bike Product Cost per Unit = \$510.73

B. Racing Bike Product Cost per Unit = \$739.61

Calculation of unit product costs:

Machining Rate = $\$22,660,000 \div (262,500 + 277,000) = \42.002 per machine hour
Engineering Rate = $\$8,652,000 \div (79,000 + 57,000) = \63.618 per engineering hour
Packing Rate = $\$3,304,000 \div (57,500 + 57,500) = \18.154 per packing order

C.

	Mountain	Racing
Machining:		
\$42.002 × 262,500	\$11,025,525	
\$42.002 × 277,000		\$11,634,554
Engineering:		
\$63.618 × 79,000	5,025,822	
\$63.618 × 57,000		3,626,226
Packing:		
\$18.154 × 57,500	1,043,855	
\$18.154 × 124,500		2,260,173
Total overhead	\$17,095,202	\$17,520,953
Units	÷ 281,500	÷ 125,500
Overhead per unit	\$60.73	\$139.61
Prime cost per unit	450.00	600.00
Unit cost	\$510.73	\$739.61

C. Mountain Bike Product Cost per Unit = \$510.73 × 1.40 = \$715

Racing Bike Product Cost per Unit = \$739.61 × 1.35 = \$998

Solution

Armstrong Company produces a variety of bicycles. One of its plants produces two bicycles: a mountain model and a racing model. At the beginning of the year, the following data were prepared for this plant:

	Mountain	Racing
Quantity	281,500	125,500
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Armstrong Company uses activity-based costing to calculate product costs.

Required:

If required, round each activity driver to the nearest cent. Use rounded amounts in subsequent computations, if applicable.

A. Calculate the per-unit product cost for a mountain bike.

\$

B. Calculate the per-unit product cost for a racing bike.

\$ 739.61

C. Assuming Armstrong Company adds 40% to the cost of a mountain bike and 35% to the cost of a racing bike, calculate the selling prices for a mountain bike and a racing bike. **Round "Unit product costs" calculations to 3 decimal places and round your final answers to the nearest whole dollar.**

Mountain Bike \$ 715

Racing Bike \$ 998